Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Roach, Pat R		
Project Title:	A Microgravity Helium Dilution Cooler		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-3191
Organization Name:	NASA Ames Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
Start Date:	02/01/1998	**End Date:**	02/01/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	2.11		
Performance Goal Text:	Adapt precision measurement and control technologies developed by fundamental physics research activities to solve needs of the human and robotic exploration of space		
Task Description:	A dilution cooler that operates well using liquid helium on the ground can be modified to operate in microgravity by adding a matrix of fine pores to the liquid chambers so that the liquid helium is held in place by capillary forces. The pores must be small enough to hold the liquid properly but large enough to allow the flow required for the operation of the cooler. To develop a continuously- operating helium dilution cooler that can reach temperatures below 0.1 K in the microgravity of space. Unique physics experiments become possible when the microgravity environment of space is combined with very low temperatures. The dilution cooler being developed can be used to precool an adiabatic demagnetization system capable		

	of reaching temperatures below 0.002 K; then it is possible to perform experiments on superfluid 3He. This liquid is one of the most unusual substances ever discovered. A number of very interesting experiments have been suggested that use the microgravity of space to allow superfluid 3He to float free of the influence of its container. In addition, many space telescopes need temperatures below 0.1 K to cool their detectors for increased sensitivity and reduced noise. A continuously-operating dilution cooler is ideal for such applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)